Interactive, Visual-Learning based Tool for Hearing Impaired Children to Improve Language and Cognitive Skills

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2023 International Conference on Information Technology (ICIT) | 979-8-3503-2006-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICIT58056.2023.10225863

***Abstract*—Hearing impairment is a common condition that affects millions of people worldwide, with approximately 1 in 1000 newborns experiencing some degree of hearing loss. This condition can significantly impact a person's quality of life, causing communication difficulties and social isolation. Early childhood hearing impairment can present significant challenges to a child's cognitive development, making it difficult for them to learn in a traditional educational setting. Consequently, there is an urge for effective learning tools that can assist hearing-impaired children in learning their first language. This paper introduces visual-based and interactive learning tools as a promising approach to enhancing the learning experience and engagement of hearing-impaired children. The proposed system utilizes machine learning to assess a child's initial status and subsequently generates adaptive content while continuously monitoring progress. Moreover, the system incorporates an object exploration feature that enables children to learn from their surroundings. Additionally, it employs natural language learning processes to present contextually similar content, supported by audio and lip movement analysis features that guide correct pronunciation. In response to the ML models utilized, object detection was accomplished through YOLO5, while lip movements were analyzed using LipNet. Regression models were deployed for assessing a child's initial status and subsequent progress. The data collection process involved comprehensive sources that covered a wide array of visual, auditory, and linguistic elements, with the dataset being rigorously tested and verified. Testing implementations were conducted in various real-world settings, encompassing diverse environmental factors and learning conditions. The YOLO model demonstrated an accuracy of 90.3%, LipNet achieved 92.2% accuracy in lip movement analysis, and the multivariate regression models showed a prediction accuracy of 92.8% in evaluating child progress.**

Keywords—Hearing impaired children, Learning tool, Language development, Cognitive development. NPL, Object detection, Lip movement analysis, Regression models

I. INTRODUCTION

Hearing impairment is one of the most common conditions that affect millions of people worldwide. According to the World Health Organization (WHO), approximately 1 in 1,000 newborns have hearing loss ranging from mild to profound [1][2]. This condition significantly impacts a person's quality of life because it causes communication difficulties, potentially leading to social isolation [3]. Hearing impairment refers to a range of hearing loss, varying from mild to profound. This condition significantly impacts a child's auditory abilities, posing challenges for their learning in traditional educational settings.

The learning journey of a child begins from the moment they are born. Cognitive development during early childhood lays the foundation for future academic success and lifelong learning [4]. The early acquisition of a first language plays a crucial role in a child's overall development. Research has shown that children should ideally learn their first language by the age of five [5]. Linguistic abilities have a profound impact on memory development, attention, and problem-solving skills, all of which are directly linked to academic success [5]. In the case of hearing-impaired children, delayed language development can lead to lower academic achievement and social isolation [3].

With the significant advancement in medical technology of the past decades, there are many advanced treatment options available for hearing-impaired people [25]. Even though hearing-impaired children are fortunate to get the ability to hear with these advanced treatments, the "Poverty of the stimulus argument" by Noam Chomsky explains the condition where a lack of auditory stimulation makes mastering a language difficult for a hearing-impaired child [24]. However, as incurred by the United Nations Sustainable Development Goals (UN-SDG), the use of ICT can fill this gap effectively [23]. To this end, this paper presents a visual-based and interactive learning tool as an effective method for enhancing the learning experience and engagement of hearing-impaired children [23][25]. Compared to traditional learning tools, visual learning tools can be used for providing a more dynamic and interactive learning experience to hearing-impaired children [26].

In contrast, this study uses YOLO5 for object detection features, word embedding, and k-means clustering for similar word generation, LipNet for video-based lip detection, and the random Forest algorithm for level determination and progress evaluation.

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II. Literature Review

Language development is critical to a child's cognitive growth and communication [18]. However, hearing-impaired children struggle for learning language due to their difficulty in learning sounds and gestures, which can impact their ability to express themselves, understand others, and develop cognitive abilities like memory, reasoning, and visualization [19][20]. Hearing-impaired children's cognitive development can be adversely affected in areas like language and audible memory [23]. Therefore, alternative methods and techniques are necessary to improve their linguistic skills [11]. Language acquisition during the first 4.5 years, supported by brain plasticity, is crucial [12] and limited vocabulary exposure also impacts these children's education [13].

Language development comprises five stages as presented in [21] ranging from pre-production to advanced fluency, but hearing-impaired children may need to learn new words and improve their vocabulary to fully develop cognitive abilities. To enhance the cognitive abilities of hearing-impaired children, innovative methods like gamified language learning can be employed by allowing them to learn at their own pace, providing engaging content, and a customized learning experience [22][23][7]. Online learning tools which apply gamification techniques have proven effective in improving teaching methods, suggesting suitable materials, and enhancing learners' growth [8][9]. Gamification encouraged specific behaviors, engaged hearing-impaired children effectively, and helped them to track their progress. Such studies also articulate that these tools are particularly beneficial for hearing-impaired children who might face challenges in accessing audiology centers. With the rise of online education, these tools offer a convenient and cost-effective solution [21]. Consequently, A web-based game application in Portugal used gesture detection to teach sign language [15], and an application in India focused on teaching Indian Sign Language (ISL) [16].

A study by the Community Child Health, Royal Children's Hospital, Australia, found a link between the age of detection of hearing impairment and education outcomes [14]. The study involved 132 students aged 7-8 years who underwent a three-hour assessment to measure educational outcomes under the supervision of a speech pathologist and a psychologist. The authors found that early detection of hearing impairment is crucial for better linguistic abilities. Additionally, the study done in [12] shows brain plasticity plays a significant role in rapid language acquisition [12], whereas [13] confirms that hearing-impaired children have limited vocabulary exposure, affecting their education.

Referring to the same papers[12][13], it is evident that special education and early intervention can assist in developing language skills and cognitive abilities in hearing-impaired children. A student-centric learning system is suggested that provides an interactive environment for children to learn from their surroundings. Furthermore, some computer games have been used to aid the development of sign language skills. In the identified applications, computer games have been strategically utilized to enhance children's attention span and stimulate their cognitive development. Similarly, this paper presents a study based on the Sri Lankan education system to support and improve the cognitive and speech abilities of hearing-impaired children based on a real case study.

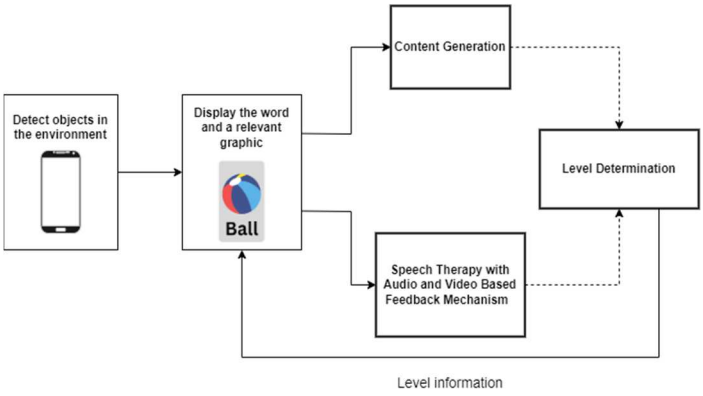


Figure 1. High-level system architecture

III. Methodology

This study divides into four sub-components as Object exploration, Similar Word Generation, Lip Movement Analysis, and Linguistic Level Determination. High-level relationships between the components are illustrated in Fig. 1.

In the process of dataset collection for the system, a combination of public datasets and locally sourced datasets from various institutes are used. Incorporating both types of datasets contribute to a more comprehensive and diverse data pool, which is instrumental in enhancing the system's effectiveness and adaptability. This approach also facilitates a robust foundation for the system's processes, enabling it to perform optimally across different tasks and settings.

Object detection uses a combination of the COCO dataset and 100 custom images from domestic environments to train the One-shot learning model and detections. In contextually similar word generation, text data was extracted from over 300 children's storybooks and 40 children's encyclopedias. This results in a corpus of 83 million tokens. The lip reading and speech therapy, relied on the GRID corpus dataset, comprising audio and visual recordings of 34 speakers producing 1,000 sentences each, resulting in a dataset of 34,000 sentences with a duration of 28 hours. The level determination utilizes 800 images and 1000 audio data collected from Twinkle Stars Pre School Rathnapura. These diverse data collection approaches provided a robust foundation for the research and its respective components.

A. Object Exploration

Object exploration, aimed at teaching the child using their surrounding environment, is the primary teaching mode for a hearing-impaired child. The process comprises two main stages, identifying the object, followed by understanding its key features. It was achieved using several machine learning models to understand the surroundings. Initially, You Only Look Once (YOLO) v3, YOLOv4, and MobileNet were evaluated for object detection through a video feed. Similar to research [31], the YOLO models were trained. The number of detections and accuracy of bounding boxes varied among the tested models. YOLOv5 outperformed the others in terms of accuracy, speed of detection (766ms), and its ability to detect small-sized objects. These models were specifically trained for indoor, and domestic environments. Child-friendly class labels were used.

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The output frames of the detection model have a complicated image of multiple detections and need to be re-processed, to avoid multiple overlapping and repetition. For the child's better understandability, the number of bounding boxes needs to be reduced, hence the number of same-class objects, IOU (Intersection Over Union), and the total number of objects within a frame are considered input parameters for classification. For the base threshold, IOU 0.6 is used to eliminate the overlapping bounding boxes.

Adding new labels to the pre-trained model was a major requirement of the system. This process was limited by the requirement of high-volume data and extensive training time. A Siamese network-based one-shot classification [27] is used to add new labels to the model. It was used to fulfill the need of training a classification model with less data. The Siamese model creates a similarity score between 0 and 1 for each image and the comparison of the base images with the given images determines the class of the image. The model was trained for certain threshold values and the best results were obtained with 38951745 trainable parameters and the input tensor was set to 105px × 105px. The Siamese model performs best in uniform backgrounds and results in overfitting on complex backgrounds.

Training models will be incorporated into an interactive, game-based environment to enhance the learning experience. The mobile device's camera will capture the environment in real time, with the YOLO object detection model identifying objects. The instance segmentation model will then segment these identified objects, amplifying the child's user experience. The game displays the name of the detected object in a scrambled form, accompanied by its pronunciation of it. The child's reaction time and spelling accuracy will be collected while the child is interacting with the gamified environment in the app.

B. Contextually Similar Word Generation

This study aims to generate contextually similar words using the gained vocabulary with object exploration. Word generation utilizes explored words from object exploration as input and produces contextually similar words. Fig. 2 depicts the process of contextually similar word generation.

Given the nature of the target audience, employing a customized dataset for model training is essential to ensure a child-friendly language model. A language model, supporting the word generation sub-process, is developed using the training corpus. Words in the corpus undergo encoding into fixed-length vectors through a positional encoding process, which conveys information about the tokens' order within a sequence. Positional encoding enables distinguishing between different token permutations. Dense, low-dimensional vector representations are resulted from this process, capturing semantic relationships between words.

Clustering the resulting embeddings into 'n' groups is accomplished using the k-means clustering technique. The goal of word embedding clustering is to identify groups of similar words based on their contextual meanings. Adapting the k-means clustering algorithm involves calculating the distance between word embeddings as a similarity metric. This approach assigns a word to a cluster where the sum of squared distances between each word embedding, and its assigned cluster centroid is minimized.

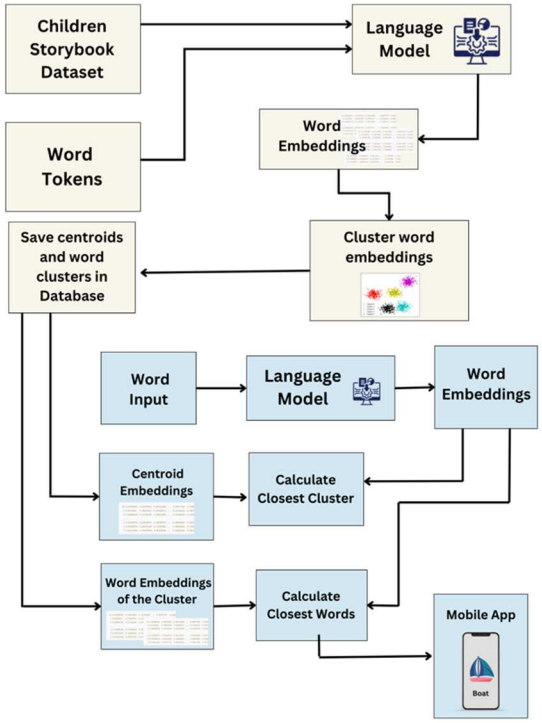


Figure 2. The flow of similar word generation

$$Minimize\ J = \sum_{i=1}^{n} \left(w_i - c_{i(j)}\right)^2 \quad (1)$$

$$suject\ to: j(i): argmin_j\ d(w_j, c_j)$$

$$where,$$

$$n\ =\ number\ of\ word\ embeddings$$

$$w_i\ =\ word\ embedding$$

$$c_{\ i(j)}\ =\ centroid\ assigned\ to\ a\ given\ w_i$$

$$j(i)\ =\ index\ of\ the\ centroid\ closest\ to\ w_i$$

The acquired vocabulary serves as input for the word-generation process. It first extracts the word embedding of the input word and subsequently determines the context class by measuring its distance against a pre-defined set of centroids. The context class containing the centroid with the least distance to the input word is considered the context class of the input word. To compute the distance, the Euclidean distance is used, which can be represented as follows.

$$d\ =\ \sqrt{\sum (x_i\ -\ y_i)^2} \quad (2)$$

The most contextually similar word is extracted from the input word's context class by computing the Euclidean distance against words in the same context class. In addition to the similarity for a given word, word complexity from the Level Determination component is also considered. The chosen contextually similar word is displayed in a gamified environment as the final step of the Contextual Similar Word Generation.

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C. Speech therapy using voice and lip-reading-based feedback mechanism

The speech therapy component involves a feedback mechanism that captures the child's pronunciation using audio and video. A composite error rate is calculated for the pronunciation correctness of the displayed word.

1) Speech detection using audio

A deep learning architecture called "Deep Speech"[29] is used for on-device word classification using audio input. First, the raw audio signal is pre-processed to extract the spectrogram features, which are then fed into the model similar to the literature [30]. The model then uses multiple layers of LSTM units to learn representations of the input features, with each layer adding additional complexity and abstraction.

To further improve the accuracy of the model, Connectionist Temporal Classification (CTC) loss is used[29]. The loss function of this algorithm allows it to handle variable-length inputs and outputs, which is crucial for accurate speech transcribing.

The RNN model exhibits a simpler structure compared to similar models in the literature [29]. This simplicity stems from limiting the design to a single recurrent layer, which is challenging to parallelize, and excluding the use of Long-Short-Term-Memory (LSTM) circuits. Employing a homogeneous model optimizes the computation of recurrent activations for maximum efficiency. This optimization involves executing only a few highly optimized BLAS(Basic Linear Algebra Subprograms) operations on the GPU and applying a single point-wise nonlinearity to compute the ReLU outputs.

2) Video-based lip-reading

Lip-reading is the main mode of speech validation when audio noise is present. The input from the camera captures the facial landmarks and the lip movements. Fig. 3 depicts the image-processing pipeline.

BlazeFace model[28] employs an enhanced network architecture derived from MobileNet. The original authors observe that the computation time for a 3×3 depth-wise convolution of a 56×56×128 tensor is only 0.07ms on the iPhone X CPU, while the subsequent 1×1 convolution from 128 to 128 channels takes 4.3 times longer at 0.3ms. Based on this, 3×3 depth-wise convolution is used, which reduces the model's depth and improves processing speed.

$$[u_t, r_t]^T = sigm(W_t z_t + W_h h_t + b_g) \quad (3)$$

where,

$z := \{z_1, \ldots, z_r\}$ *is input sequence to the RNN*



Figure 3. Image processing pipeline

For training the lip movements for pronunciation, the GRID corpus dataset [17], which includes both audio and video recordings of 34 speakers is used.

The proposed model is based on a 3D convolutional neural network that takes in video frames as input and outputs text at the sentence level. The fine-tuned model achieves good results(89.2 %) in lip-reading accuracy.

The audio component's score and the LipNet's score are individually calculated and later aggregated using a weighted average. The results were then forwarded to level determination and other modules for their use.

$$Weighted\ average = \frac{\sum w_i x_i}{\sum w_i} \quad (4)$$

D. Initial Status Determination and Progress Monitoring

Consists of two sub-components, which aim to determine the initial status of the user and monitor their progress. The initial status of the user is determined using a dynamic quiz river. The quiz river illustrated in Fig. 4 contains simple words that are divided into five levels. The accuracy of the response, response time, and the calculated difficulty score are inserted into the Random Forest algorithm to determine the user's initial status.

The progress monitoring process includes a feedback mechanism that motivates users to interact with the application. Learning materials were obtained from the CEHIC (Centre for Education of Hearing-Impaired Children) The level of difficulty is determined by the semantic difficulty and number of letters. Furthermore, this evaluates the knowledge of basic words such as three-letter words, four-letter words, five-letter words, and complex words that has more than five letters or a combination of two words. Each question is displayed along with a pronouncing voice clip of the word playing in the background.

The component uses a model implemented with the regression algorithm to determine the level of the user based on the response accuracy, response time, and final quiz difficulty. The personalized data of the user does not have linear relationships with each other, so this method can be used to evaluate the progress level of the user.

The gamified solution has a quiz that has fifteen questions in total on five different levels. Three questions will be randomly selected from each level for the quiz and those will be displayed. The child must select the accurate

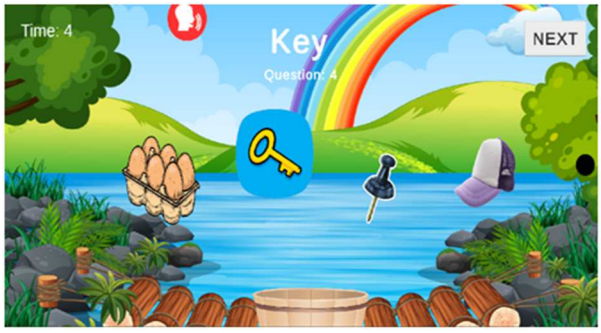


Figure 4. UI of Quiz River

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image and then drag and drop that into the basket. Fifteen seconds are given to answer one question. The audio clip pronouncing the word can be played any number of times within those fifteen seconds. If the child answers the question incorrectly, they can still try again until the fifteen seconds are over. The next question will be displayed as soon as the timer reaches zero. There is an option available to skip a question and navigate to the next question.

The dataset used for the training of the random forest algorithm consisted of the accuracy of answers given by the child for each question in the quiz. To acquire the difficulty of the question different points were given for each level of question. The points allocated to each question are equal to the difficulty level of that question. Those points are allocated if the answer was right and zero is allocated if the answer is incorrect.

The variables which are impacting heavily on the level of the child are taken according to this and then data is preprocessed and split as train and test data. In the developed solution 20% of data are taken for testing. Then the model is trained along with the hyperparameters which are n_estimator, max depth, and min_samples_leaf. N_estimator specifies the number of trees, max_depth specifies the number of splits that each tree is allowed to make and min_samples_leaf specifies the minimal number of samples required at the leaf. After the training of data, the results that give maximum accuracy for detecting the level of the user are taken and passed to get the maximum accuracy level of the child.

Parameter tuning in the random forest algorithm was cruicial in optimizing the performance and accuracy of the predictions. The key hyperparameters in random forest algorithm are number of estimators (n_estimator), maximum depth(max_depth) and minimum samples per leaf (min_leaf). Those parameters were tuned in the model under three ranges and hyperparameter values were tuned as 20, 20 and 1 for n_estimator, max_depth and min_samples_leaf respectively to finalize a model with highest accuracy.

V Results and Discussion

The model for Object Exploration primarily utilizes YOLO v5 architecture trained with the publicly available COCO dataset and 100 captured images from the domestic environment. 766ms of on-device inference time is achieved with model compression and fine-tuning. During optimizations, an ideal input size of 1280×1280 was identified. Siamese network-based one-shot classification is used for updating pre-trained model weights to classify new labels. It demonstrated accuracies of over 89% under different environments/backgrounds. The limitations of this technique were complex backgrounds make the model overfit for the images. Using the trial-and-error method, parameters were adjusted, and the best results were obtained with 38951745 trainable parameters and the input tensor was set to 105px × 105px.

The similar word generation mechanism was developed that generates words based on the acquired words from an object detection process. Word embeddings facilitate capturing of the contextual and semantic properties of words. These embeddings offer a lower-dimensional semantic space for high-dimensional word vectors, enabling accurate calculation of semantic distances between words. K-means clustering partitions this semantic space into distinct clusters based on the semantic relatedness of words. This process paves the way for computing the closest word to a given word based on the distance between the word embedding of the given word and the pre-classified word classes using k-means.

Despite the promise, potential limitations of the approach include the assumptions made by k-means clustering and the possible inadequacy of word embeddings in handling words with multiple meanings or varied contextual usage. Future exploration could delve into advanced clustering techniques and different word embedding types to mitigate these limitations. The research presents a significant stride in linking image processing and natural language processing.

The initial Status Determination component uses a model based on the random forest algorithm which is trained using a dynamic dataset collected with the usage of Quiz River. Datapoints include 18 different features and thousands of records. The random forest algorithm tunes the most important hyperparameters which are n_estimator, max_depth, and min_sample_leaf and comes up with a level with 91.8% of accuracy. The usability of model and the has been validated at CEHIC to check whether the model predicts the accurate level for the student based on the child's performance at the quiz river. During the validation, a correlation matrix is used to analyze the correlation of variables as illustrated in Fig. 5.

The proposed system for personalized speech therapy employs multiple models, "Deep Search" for audio-based phoneme classification which was optimized for on-device classifications. The accuracy of the model was limited by the variable length inputs, later it was improved to an accuracy of 93.4% on a custom-made test dataset, using the CTC loss function.

For the lip-analyzing vision pipeline, Blzaeface model with pre-trained weights is used for near-real-time face detection. During the testing on a mobile GPU, <1ms detection time is achieved using 3×3 depth-wise convolution. The cropped frame is then fed to lip analyzer network which uses the "LipNet" architecture. The fine-tuned model achieves an accuracy of 89.2% for inferencing the pronunciation using the spatial movement of lips. It was observed that better performance can be achieved with sentence-level prediction.

The system's adaptability caters to diverse user needs and learning styles, with the gamified approach potentially enhancing learning outcomes. However, acknowledging limitations, future research should focus on validating the system's effectiveness, incorporating user feedback, and exploring emerging technologies, such as augmented reality and haptic feedback. In conclusion, the proposed system

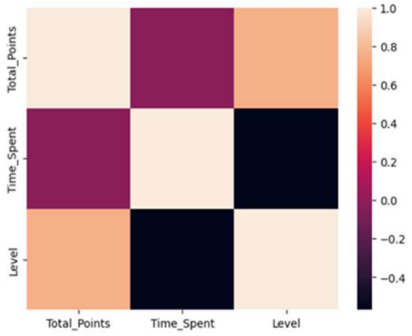


Figure 5. Coefficient Correlation Matrix

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presents a promising alternative to traditional speech therapy methods, contributing to a personalized and engaging learning experience. Further validation and user feedback integration could significantly impact speech therapy and improve hearing-impaired children's lives.

VI. Conclusion and Future Work

Using a mobile app with gamified tasks and explorative learning this paper proposes a method that can improve language and cognitive abilities of a hearing-impaired child. It was observed that all the models perform with performance with better than 84% accuracy. The application should be used in the field for many years to gather its real impact on the students. The above research work is done based on the English language and further extension can be taken to do enhanced visual learning in other languages. Currently, the models are compatible with mobile versions and further enhancement related to increasing the compatibility of mobile phones can be done. The learning process is currently done with the supervision of a teacher on premises but further improvements can be made to accommodate remote learning.

VII. Acknowledgment

We would like to thank Rev. Sr. Greta Nalawatta and her team from the Centre for Education of Hearing-Impaired Children (CEHIC) for their insightful comments and recommendations. Her expertise in the field has been instrumental in shaping the direction of our research.

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